Chemistry - Graduate Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Elements with partially filled 4 f or 5 f orbitals include all of the following EXCEPT
   (a) Eu
   (b) Gd
   (c) Am
   (d) Cu

2. Assuming complete neutralization, calculate the number of milliliters of 0.025 M H_3 PO_4 required to neutralize 25 ml of 0.030 M Ca(OH)_2.
   (a) 10 ml
   (b) 30 ml
   (c) 55 ml
   (d) 40 ml
   (e) 25 ml

3. A wine has an acetic acid (CH_3 COOH, 60 g/formula weight) content of 0.66% by weight. If the density of the wine is 1.11 g/ml, what is the formality of the acid?
   (a) $1.6 \times 10^{-}3F$
   (a) $1.5 \times 10^{-}3F$
   (a) $1.4 \times 10^{-}3F$
   (a) $0.8 \times 10^{-}3F$
   (a) $0.9 \times 10^{-}3F$

4. Under standard temperature and pressure conditions, compare the relative rates at which inert gases,Ar, He, and Kr diffuse through a common orifice.
   (a) .1002 : .3002 : .4002
   (b) .3582 : .4582 : .0582
   (c) .2582 : .4998 : .3092
   (d) .1582 : .6008 : .2092
   (e) .1582 : .4998 : .1092

5. Would you expect He_2$^{\oplus} tobemorestablethanHe_2?ThanHe$?

6. He_2$^{\oplus}isthemoststableamongHe, He_2, andHe_2^{\oplus}$

6. He_2$^{\oplus}ismorestablethanHe_2butlessstablethanHe$

6. He, He_2, and He_2$^{\oplus}havethesamestability$

6. He_2$^{\oplus}istheleaststableamongHe, He_2, andHe_2^{\oplus}$

6. He and He_2 have the same stability, both more stable than He_2$^{\oplus}$

True / False (10 marks)

Answer True or False

6. True or False: The ionization energy of singly ionized helium, calculated using the Bohr theory, is 48.359 eV.

7. True or False: At$_{12}$.19 K, a pure sample of an ideal gas can exhibit a pressure of 1 atm and a concentration of 1 mole liter^{-}1.

7. True or False: Zn(NO 3)2 is expected to have a lower solubility in water than NiSO 3 due to its more complex structure.

8. True or False: The missing mass in ^{1}9_9$Fissignificantlygreaterthanthemassofanelectron, approxi$

8. True or False: The molality of a solution with 49 g of H_2SO_4 in 250 g of water is 1.0 moles/kg.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the amount of water needed to dilute 5 gallons of a 90% ammonia solution to achieve a 60% concentration, and explain the reasoning behind your calculations.

12. Discuss the thermodynamic principles governing the expansion of one cubic foot of air when heated from $0^{t}o546,000^{\cdot}andanalyzetheresultingchangeinvolume.$

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